

AUTOMATED GREENHOUSE

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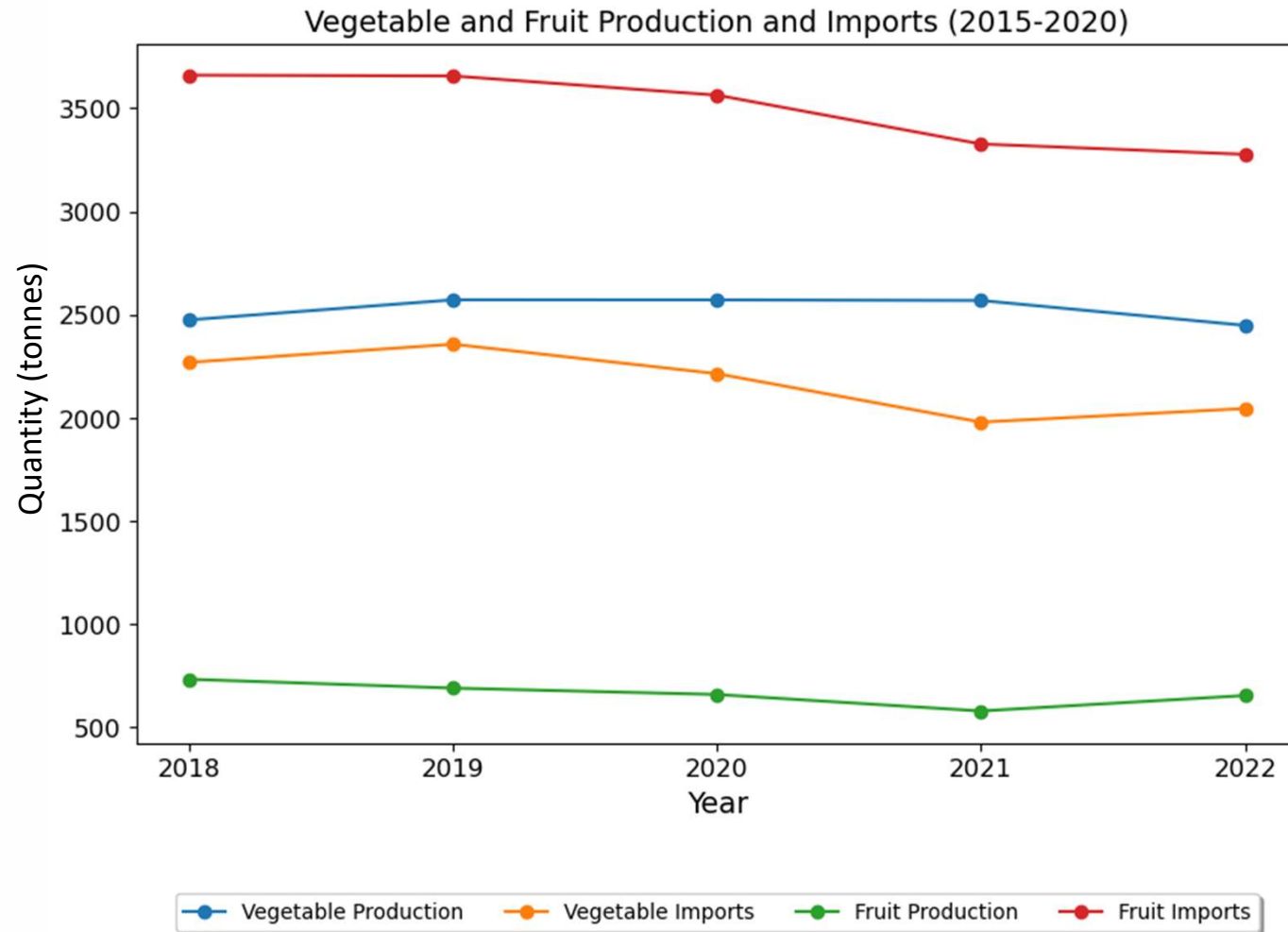
1. Motivation
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Why?



LSHTM. (n.d.). One third of UK fruit and vegetables are imported from climate-vulnerable countries – and this is on the rise. [online] Available at: <https://www.lshtm.ac.uk/newsevents/news/2020/one-third-uk-fruit-and-vegetables-are-imported-climate-vulnerable-countries>.

FOOD IMPORTS



GOV.UK. (n.d.). Horticulture statistics - 2022. [online] Available at: <https://www.gov.uk/government/statistics/latest-horticulture-statistics/horticulture-statistics-2022>.

THE METHOD

Location & Greenhouse

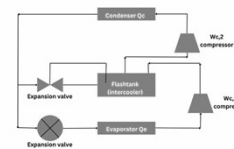


Heat transfer

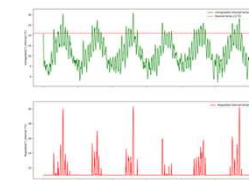
$$T_{int} (^{\circ}C) = T_{current} \times \left(\frac{Q_{total}}{mc_p} \right)$$

$$Q (W) = kWh \times T_{transmittance} \times \frac{1000}{24}$$

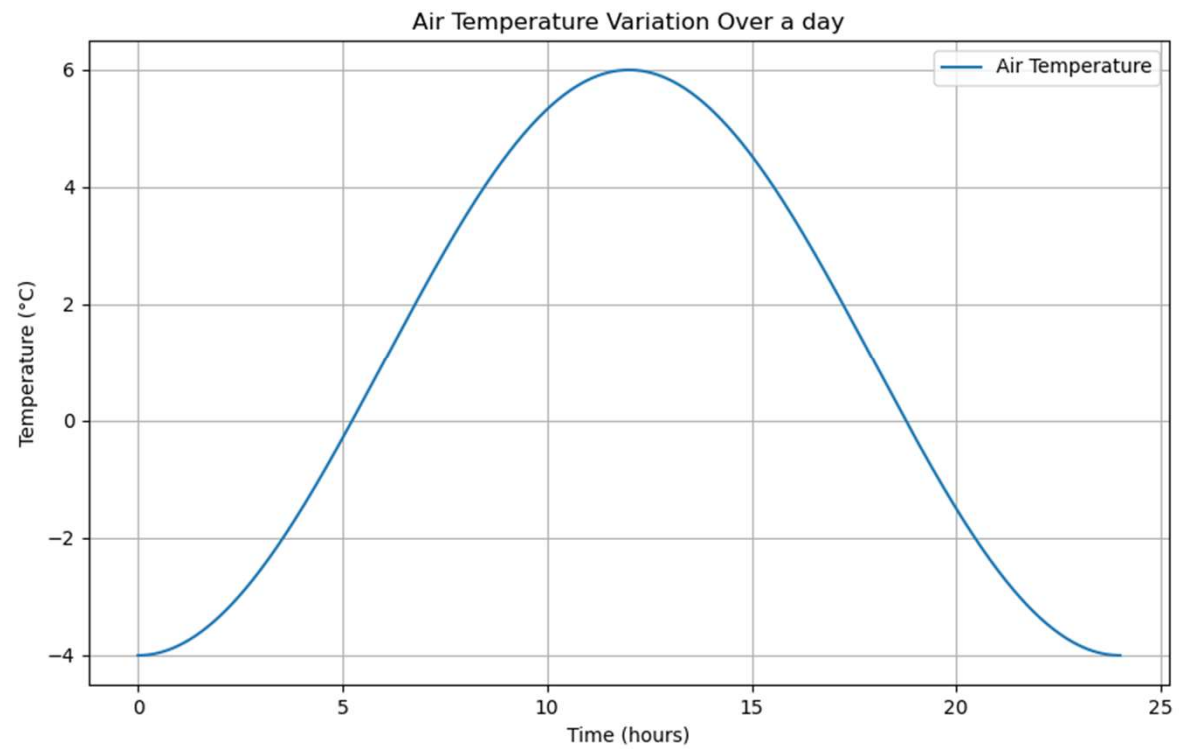
Heat pump design



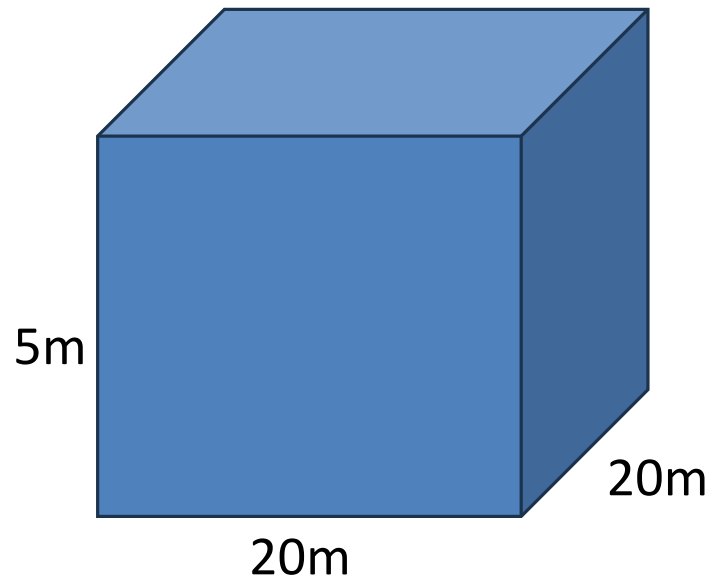
Results



THE SITE



THE GREENHOUSE



Walls/Roof:

PVC

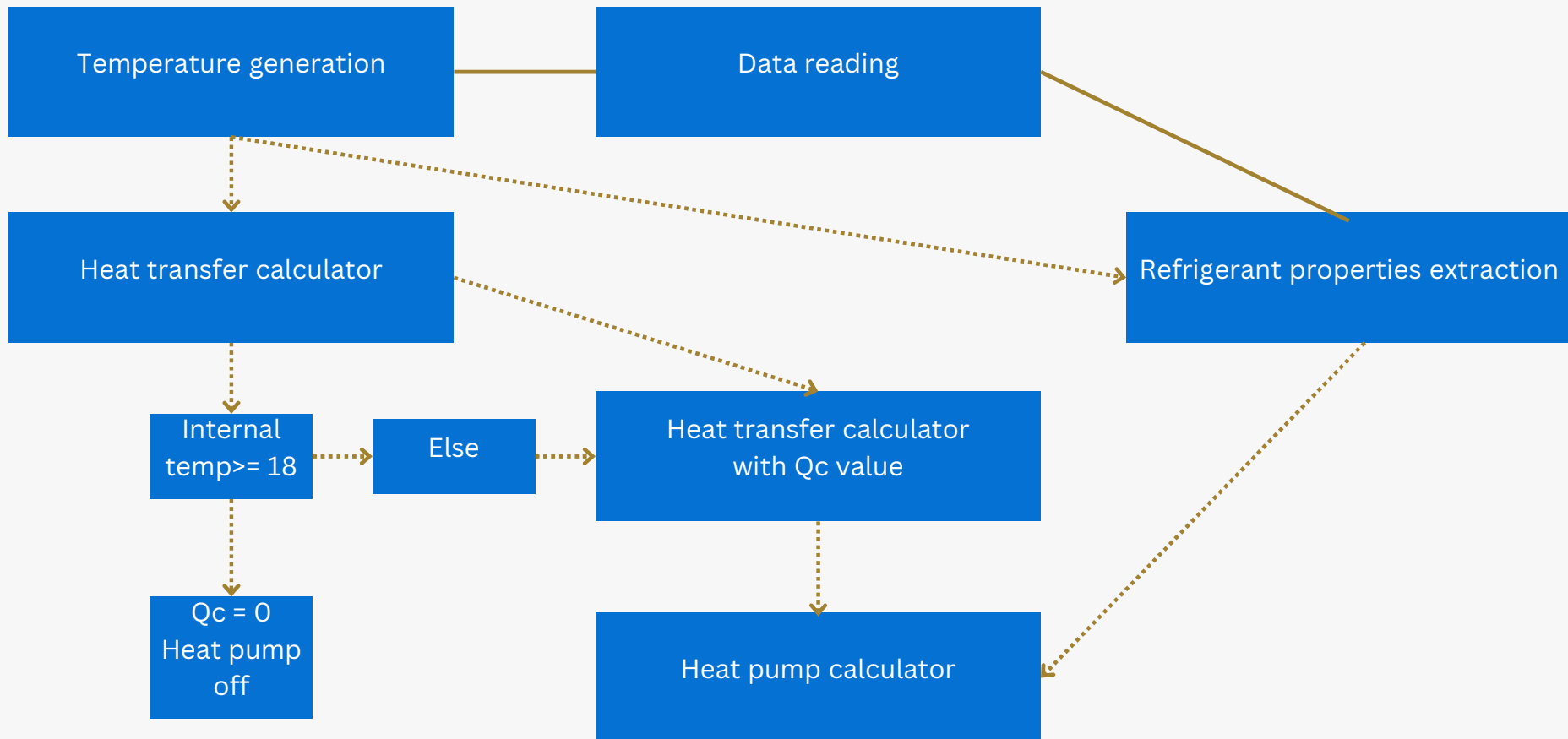
Thermal conduction
coefficient: 0.12 W/mK

Ground:

Rolled rubber

Thermal
conduction coefficient:
 0.17 W/mK

SIMULATION: UML DIAGRAM



ASSUMPTIONS

Greenhouse

- Heat dissipates evenly
- Constant pressure
- Air flow is evenly distributed, air density distribution is uniform
- PVC is in steady state
- 100% of solar kWh absorbed by panels

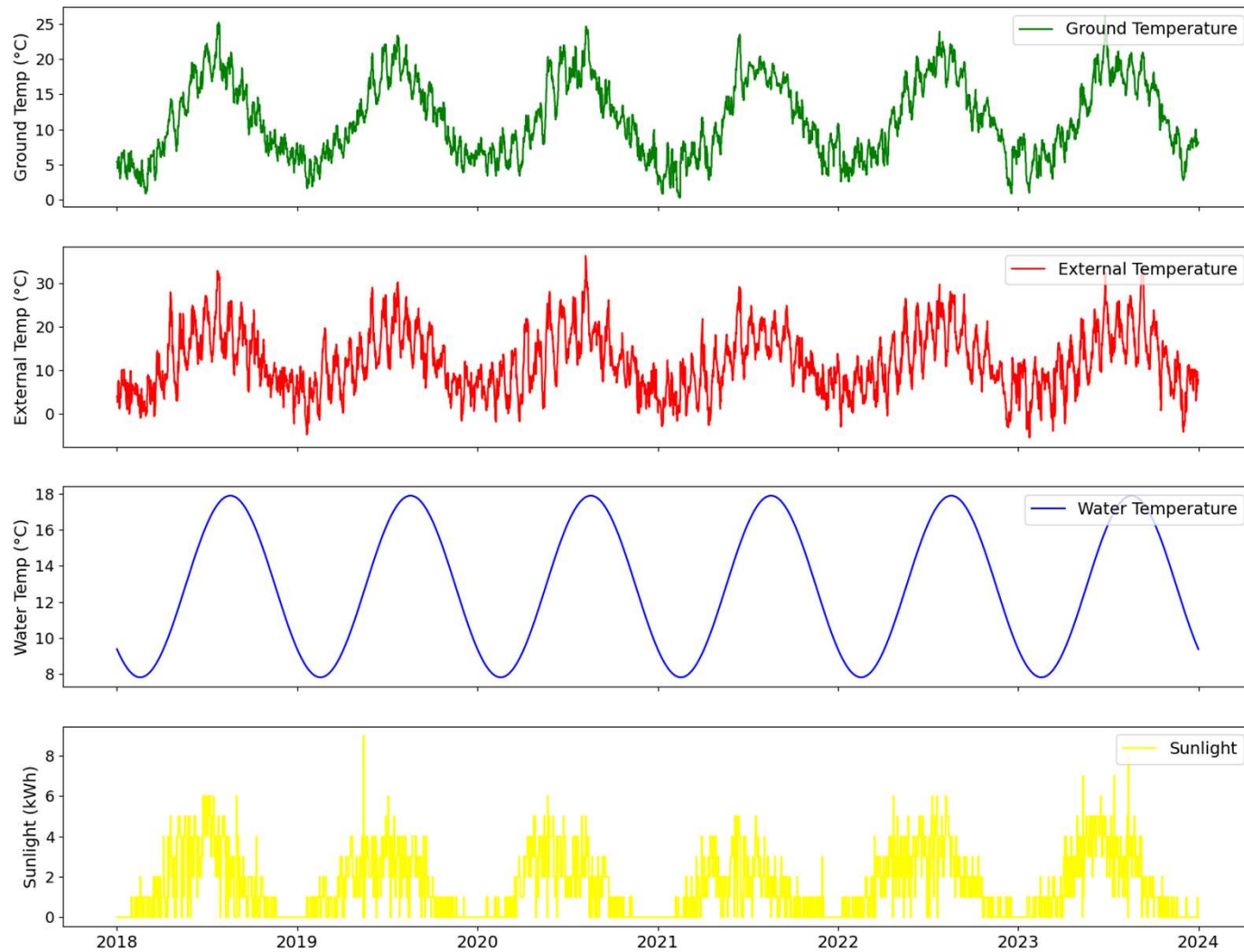
Heat pump

- Condenser losses are negligible
- Refrigerant is saturated
- Compression is isentropic

EXTERNAL CONDITIONS

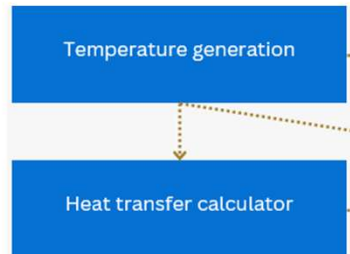
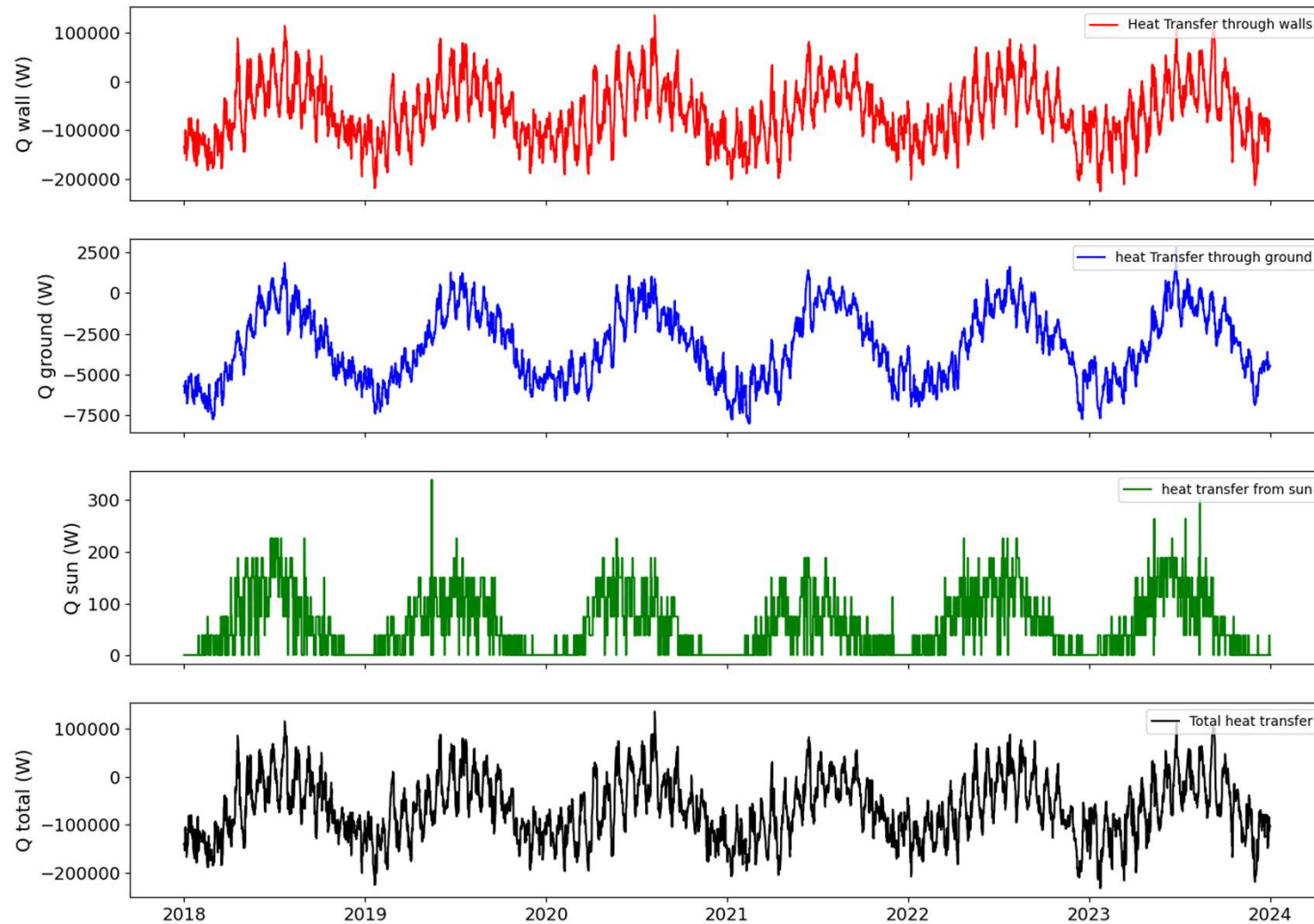
Temperature generation

Data reading

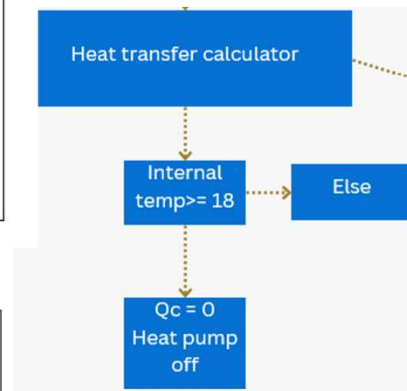
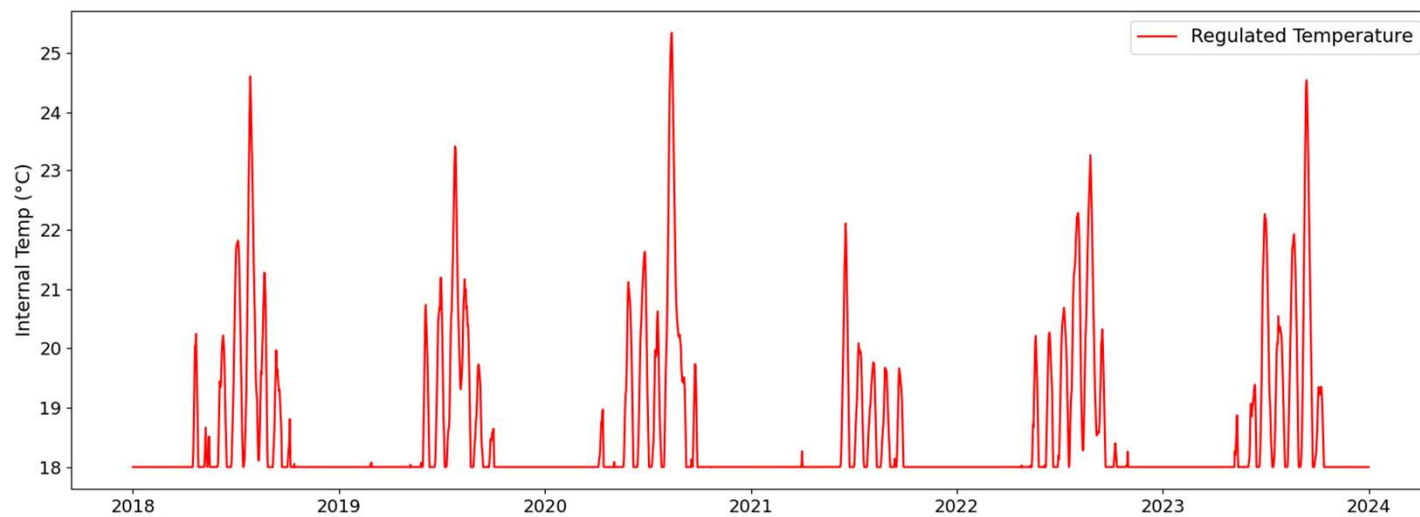
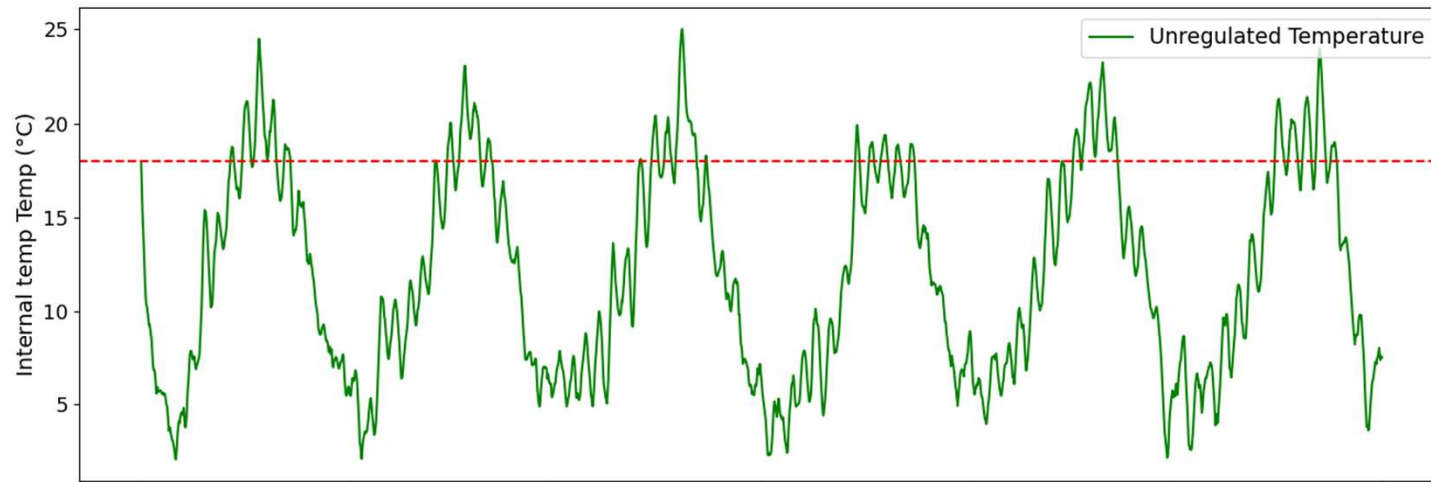


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(n.d.). Chelmsford Weather Station -
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? [Accessed 14 Mar. 2024].

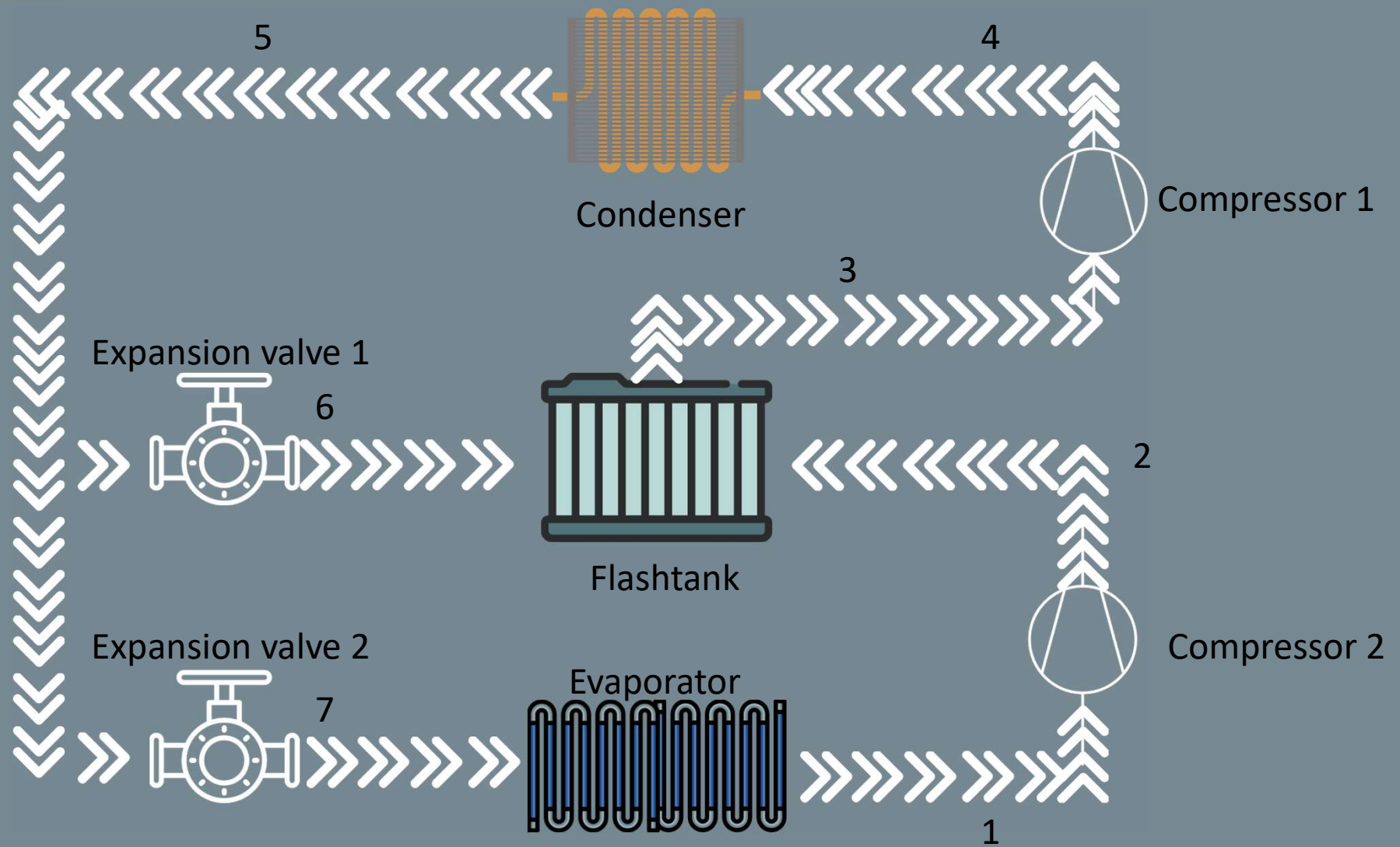
HEAT TRANSFER



INTERNAL TEMPERATURES



THE HEAT PUMP



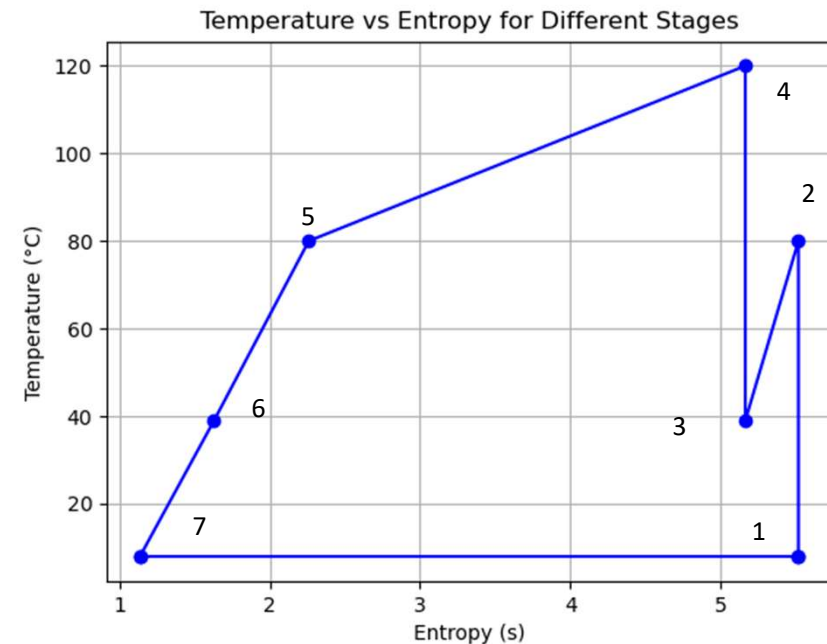
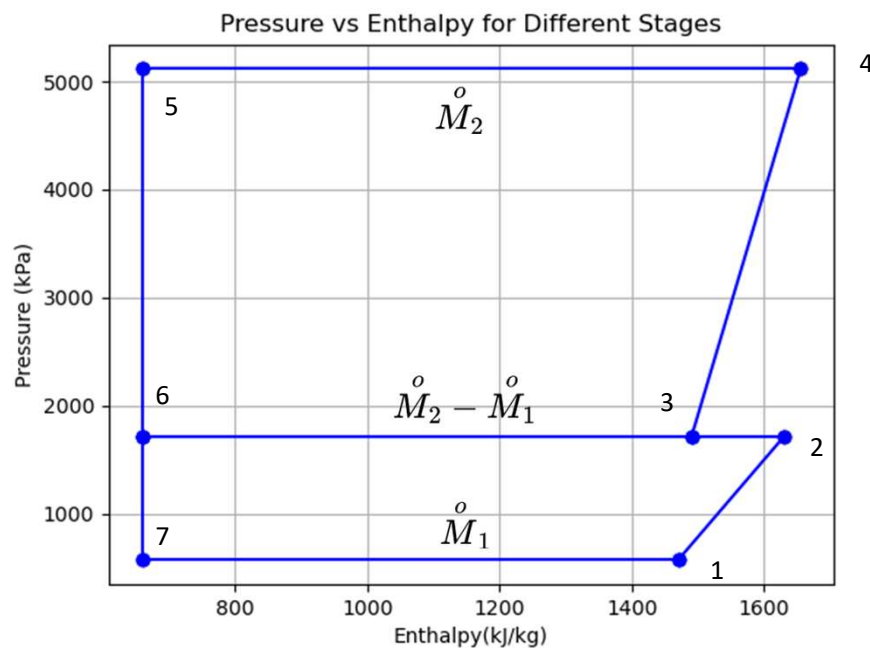
REFRIGERANT: AMMONIA

Global Warming Potential: 0

Ozone Depletion Potential: 0

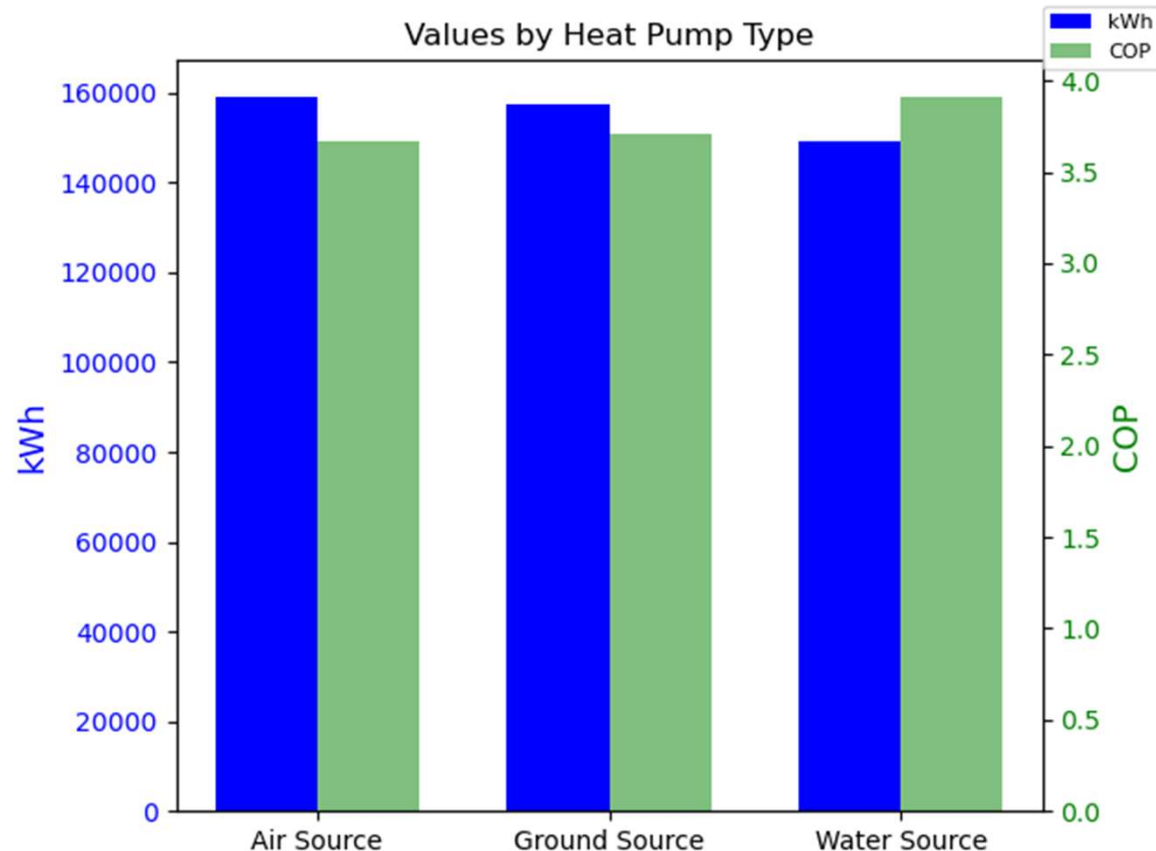
Specific heat capacity: 1343-9312 kg/kJ^oC

Refrigerant properties extraction



AN INTERNATIONAL ORGANIZATION ASHRAE Position Document on AMMONIA AS A REFRIGERANT. (n.d.). Available at:
https://www.epa.gov/sites/default/files/documents/ASHRAE_PD_Ammonia_Refrigerant_2010_1.pdf.

HEATPUMP COMPARISON



Air source

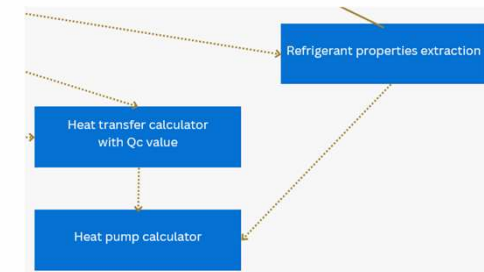
- COP: 3.66
- kwh/annum: 159,000

Ground source

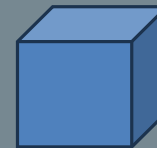
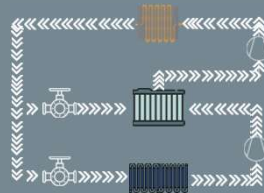
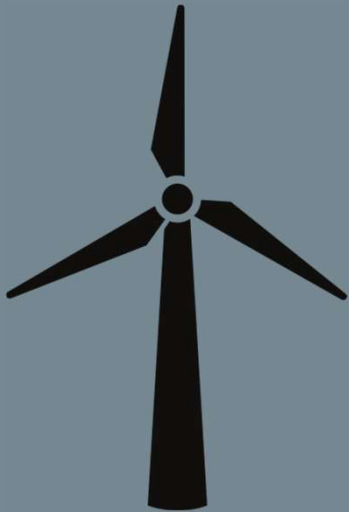
- COP: 3.7
- kwh/annum: 157,500

Water source

- COP: 3.91
- kwh/annum: 149,000



POWER SUPPLY



IMPACT



Blaauw, Sheldon and Broekman, Andre and Maina, James and JvdM Steyn, Wynand and Haddad, Wiam, Life Cycle Assessment of an Avocado: Grown in South Africa – Enjoyed in Europe. Available at SSRN: [Add a little bit of body text](#) or [Add a little bit of body text](#)

Conclusions

- Saves 700kg of CO2 emissions per tonnes of avocado's produced
- Would need 11 square kilometers to produce UK avocado demand
- Water source Heat pump is most efficient best for long term savings
- Short term costs a priority air source would be best
- Further work needed

THANK YOU

APPENDIX: EQUATIONS

Temp generation

$$T_{Ground,air} = \left(\frac{(T_{max} + T_{min})}{2} \right) + \left(\frac{(T_{max} - T_{min})}{2} \right) \times \sin \left(\frac{2\pi t - \pi}{2} \right)$$

$$T_{water} = \left(\frac{(T_{max} + T_{min})}{2} \right) + \left(\frac{(T_{max} - T_{min})}{2} \right) \times \sin \left(\frac{2\pi t}{t + 0.4\pi} \right)$$

Heat transfer

$$Q = -k \times A \times \frac{(T_{Int} - T_{Ext})}{d}$$

$$Q_{sun} = E_{sun} \times Transparency$$

$$T_{Int} = T_{Initial} + \frac{Q}{m_{air} \times c_{p,air}} + \left(\frac{Q_C}{m_{air} \times c_{p,air}} \right)$$

$$Q_c = (T_{Des} - T_{Int}) \times m_{air} \times c_{p,air} \quad \text{(With heat pump)}$$

Heat Pump

$$\dot{M}_2 = \frac{Q_c}{(h_4 - h_5)}$$

$$\dot{M}_1 = \dot{M}_2 \times \frac{(h_3 - h_6)}{(h_2 - h_6)}$$

$$Q_e = \dot{M}_1 \times (h_1 - h_7)$$

$$W_{c,1} = \dot{M}_1 \times (h_2 - h_1)$$

$$W_{c,2} = \dot{M}_2 \times (h_4 - h_3)$$

$$COP = \frac{Q_c}{W_{c,1} + W_{c,2}}$$

APPENDIX: AMMONIA EXTRACTION

Stage 1: $T_1 = T_{ground,air,water} \longrightarrow$ Extract row from saturated properties (vapor)

Stage 5: $T_5 = 80 \longrightarrow$ Extract row from saturated properties (liquid)

Stage 3: $P_3 = \sqrt{P_1 \times P_5} \longrightarrow$ Extract row from saturated properties (vapor)

Stage 6: $P_6 = P_3$ $h_6 = h_5$ $T_6 = T_3$ $s_6 = s_{3(liquid)}$

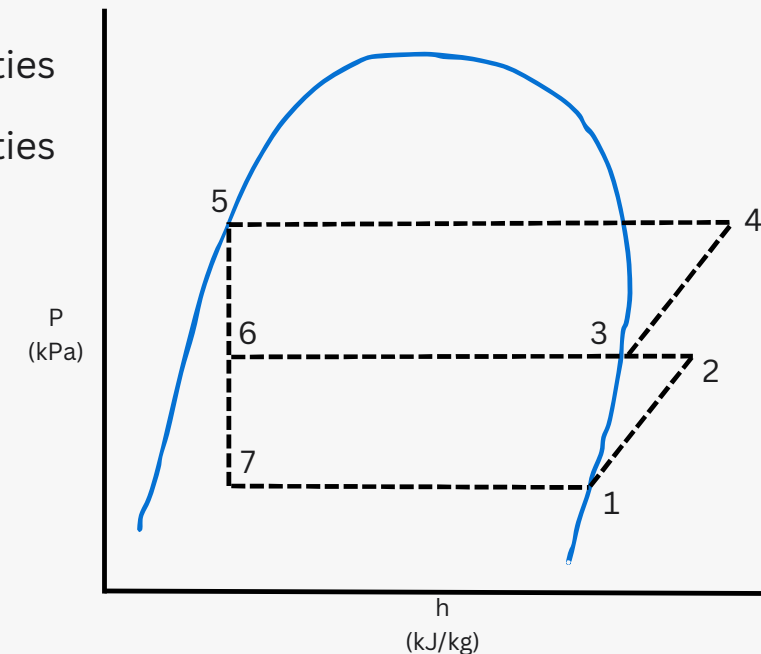
Stage 7: $P_7 = P_1$ $h_7 = h_5$ $T_7 = T_1$ $s_7 = s_{1(liquid)}$

Stage 4: $P_4 = P_5$ $s_4 = s_3 \longrightarrow$ Extract row from Superheated properties

Stage 2: $P_2 = P_3$ $s_2 = s_1 \longrightarrow$ Extract row from Superheated properties

Temp. °C	Saturation t = 5 C, p = 515.75 kPa Enthalpy kJ/kg	Entropy kJ/(kg-K)	Sp. Vol. L/kg
5	1467.393	5.5568	243.0391
10	1480.980	5.6052	249.0486
15	1494.216	5.6515	254.9226
20	1507.162	5.6961	260.6818
25	1519.869	5.7391	266.3430
30	1532.378	5.7807	271.9198
35	1544.721	5.8210	277.4232
40	1556.927	5.8603	282.8625
45	1569.019	5.8986	288.2453
50	1581.016	5.9361	293.5783
55	1592.936	5.9727	298.8669
60	1604.793	6.0085	304.1160

Temperature (°C)	Pressure (kPa)	Enthalpy (kJ/kg)		Entropy (kJ/kg-K)	
		Liquid	Vapor	Liquid	Vapor
13	681.35	260.659	1474.728	1.2155	5.4583
14	704.63	265.369	1475.562	1.2318	5.4463
15	728.52	270.087	1476.376	1.2481	5.4344
16	753.03	274.812	1477.172	1.2643	5.4226
17	778.17	279.544	1477.948	1.2805	5.4108
18	803.95	284.283	1478.704	1.2967	5.3991
19	830.38	289.029	1479.441	1.3128	5.3874
20	857.48	293.783	1480.157	1.3289	5.3759
21	885.24	298.544	1480.853	1.3449	5.3643
22	913.69	303.313	1481.529	1.3610	5.3529



https://web.iiar.org/membersonly/PDF/CO/databook_AppA.pdf

APPENDIX: RADIATOR

$$Nu = C \times (Gr \times Pr)^n = \frac{hL}{k}$$

$$Gr = \frac{g\beta \times \Delta T \times L^3}{\nu^2}$$

$$Q_{conduction} = m_{water} c_{p,water} \times \Delta T$$

$$Q_{convection} = hA\Delta T$$

APPENDIX: CONSTANTS

Air:

$$c_{p,air} = 1006 \left(\frac{J}{kg^{\circ}C} \right)$$

$$m_{air} = V \times d = 2586kg$$

Turbine:

$$K = 3.2 \quad V_{wind,avg,yearly} = 3.9$$

Radiator

$$c_{p,water} = 4200 \left(\frac{J}{kg^{\circ}C} \right)$$

$$\beta_{air} = 0.0034 (K^{-1})$$

$$v_{air} = 1.48 \times 10^{-5} \frac{m^2}{s}$$

$$Pr = 0.7$$

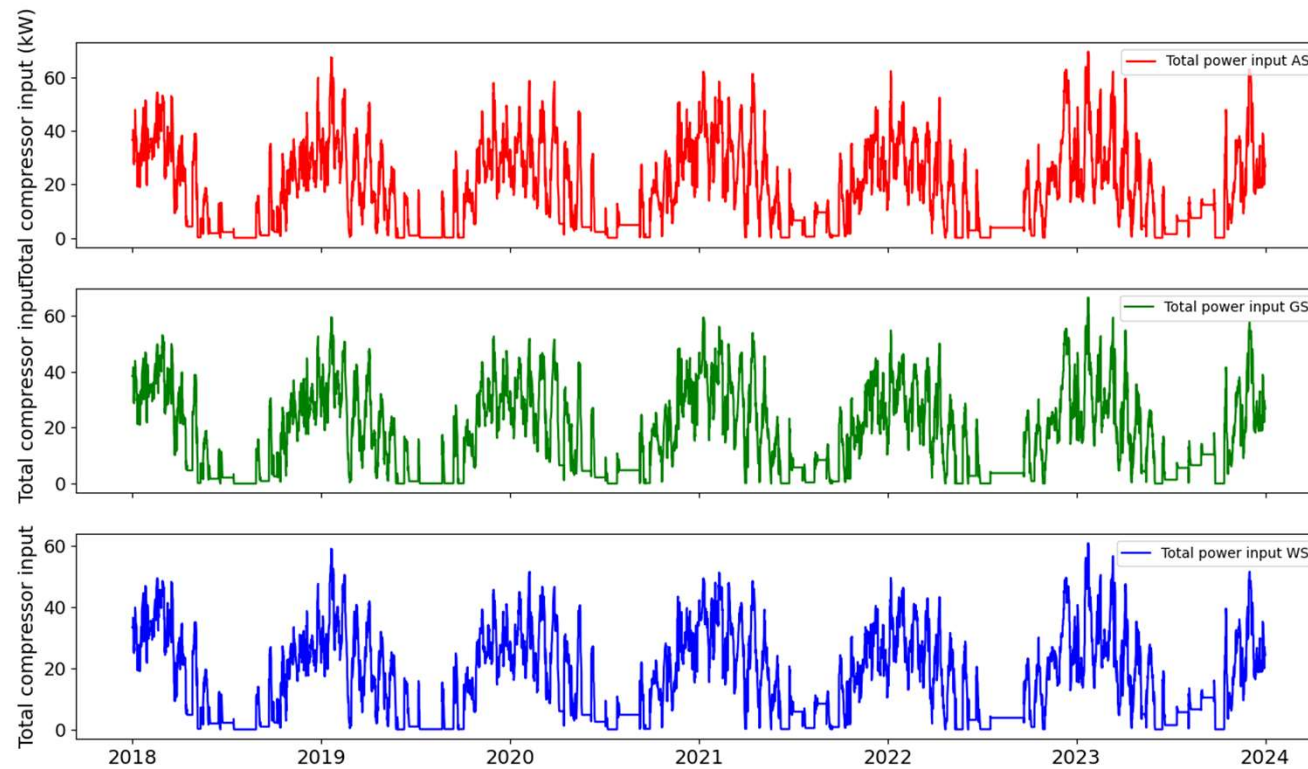
$$k_{air} = 0.003 (Wm^{-1}K^{-1})$$

Nusselt

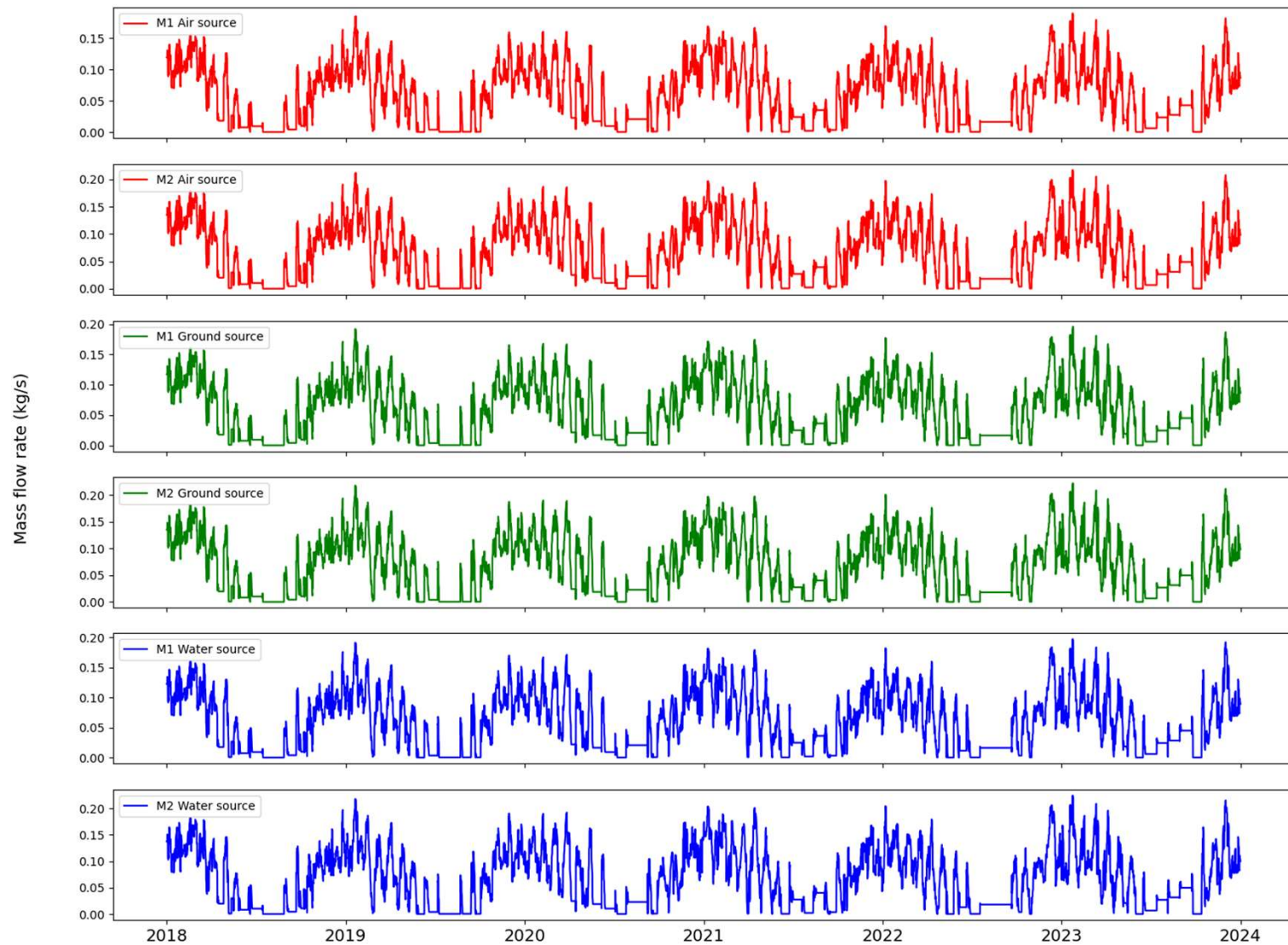
$$C_{number} = 0.59$$

$$n = 0.25$$

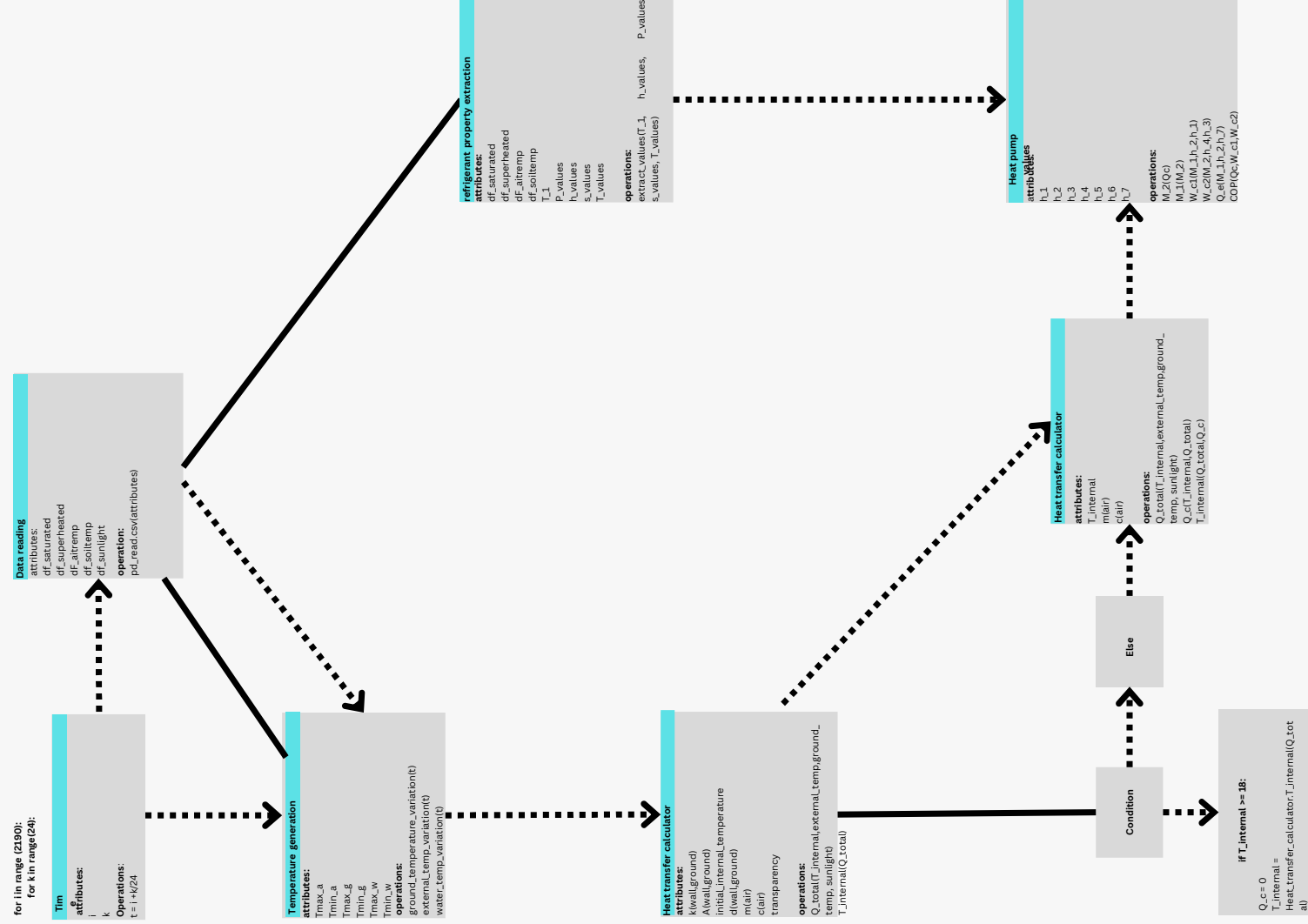
APPENDIX: COMPRESSOR POWER



APPENDIX: MASS FLOW RATES



APPENDIX: UML



ADDITIONAL REFERENCES

<https://www.geturbanleaf.com/blogs/fruit/growing-avocado-indoors#:~:text=In%20order%20to%20provide%20an,500%20%CE%BCmol%2Fm%C2%B2%2Fs.>
conversion: <https://www.waveformlighting.com/horticulture/convert-ppfd-to-lux-online-calculator>